

Fundamentals of Linear Control:

A Concise Approach

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Chapter 2: Dynamic Systems

linearcontrol.info/fundamentals

2.1 Dynamic Models

Longitudinal car model

Newton's law

$$m \dot{v}(t) + b v(t) = f(t)$$

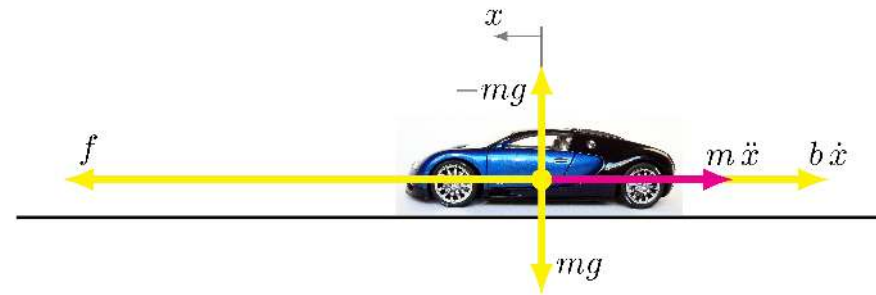
Force model

$$f(t) = p u(t)$$

Ordinary differential equation

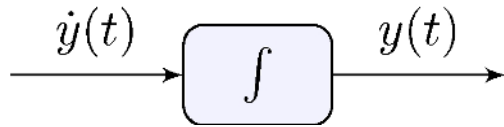
$$\dot{y}(t) + \frac{b}{m} y(t) = \frac{p}{m} u(t)$$

- input: u is pedal excursion in inches
- output: y is car velocity in mph

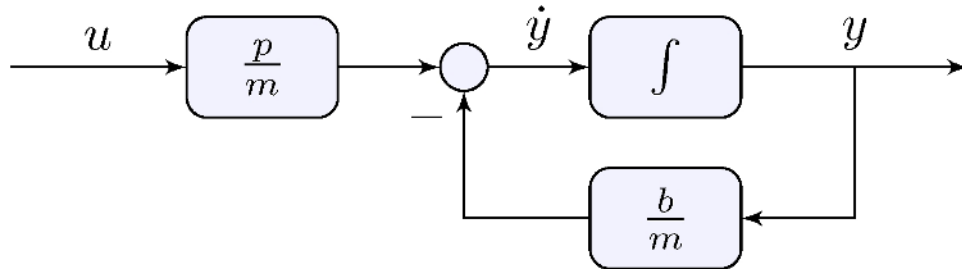


2.2 Block-Diagrams for Differential Equations

Basic element: integrator



Longitudinal car model



$$\dot{y}(t) = \frac{p}{m}u(t) - \frac{b}{m}y(t)$$

Note the presence of feedback!

2.3 Dynamic Response

Longitudinal car model

$$\dot{y}(t) + \frac{b}{m} y(t) = \frac{p}{m} u(t)$$

Constant pedal input

$$u(t) = \tilde{u}, \quad t \geq 0$$

Particular solution

$$y_P(t) = \tilde{y} = \left(\frac{p}{m} \bigg/ \frac{b}{m} \right) \tilde{u} = \frac{p}{b} \tilde{u}$$

Longitudinal car model: homogenous equation

$$\dot{y}(t) + \frac{b}{m} y(t) = 0$$

Homogenous solution

$$y_H(t) = e^{\lambda t}$$

Characteristic equation

$$\dot{y}_H(t) + \frac{b}{m} y_H(t) = \left(\lambda + \frac{b}{m} \right) e^{\lambda t} = 0 \quad \implies \quad \lambda + \frac{b}{m} = 0$$

Longitudinal car model with constant pedal input

$$\dot{y}(t) + \frac{b}{m} y(t) = \frac{p}{m} \tilde{u}$$

Particular + homogenous solution

$$y(t) = y_P(t) + \beta y_H(t) = \tilde{y} + \beta e^{\lambda t}$$

Initial conditions

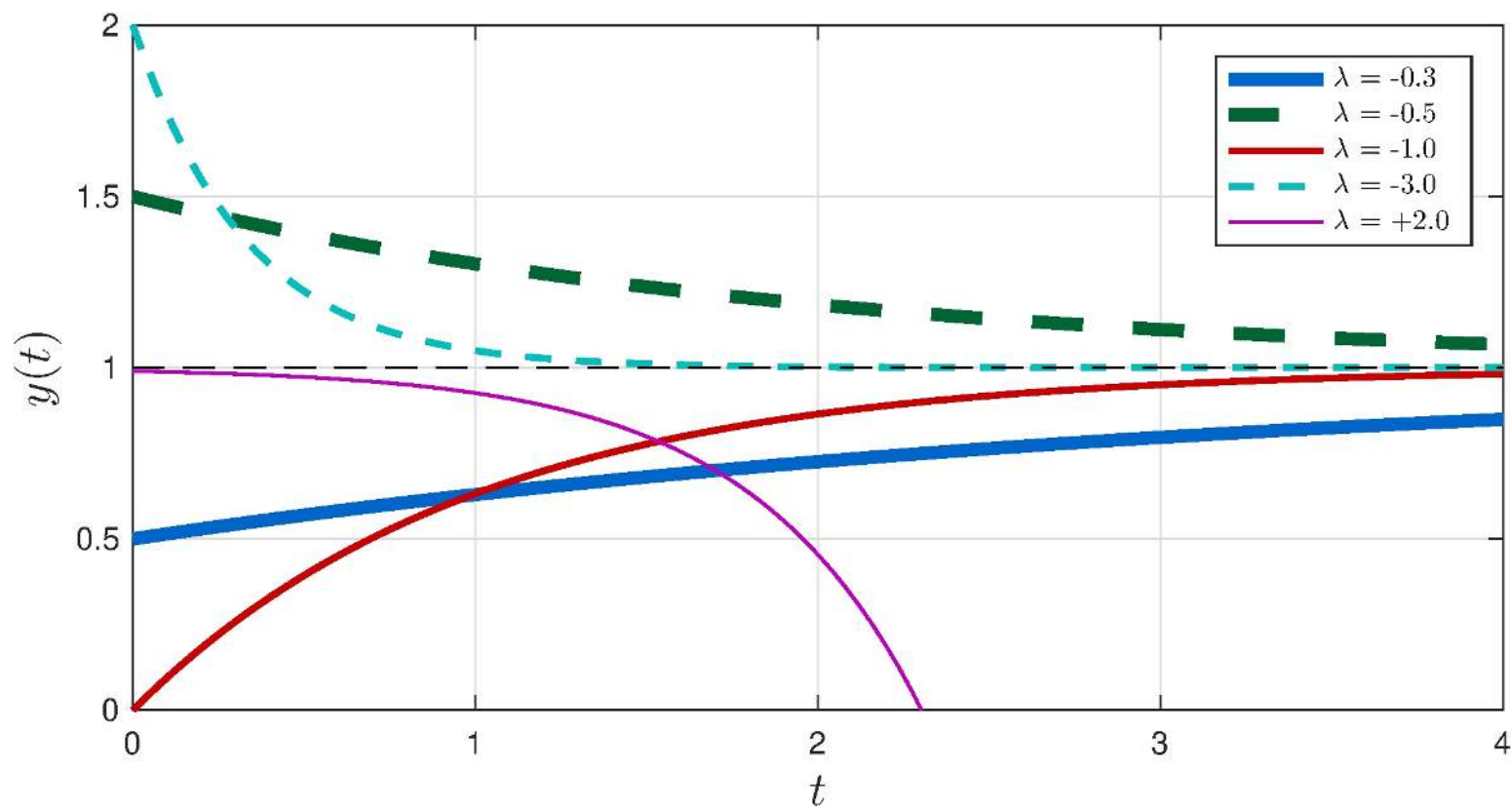
$$y(0) = \tilde{y} + \beta = y_0 \quad \implies \quad \beta = y_0 - \tilde{y}$$

Complete solution

$$y(t) = \tilde{y} (1 - e^{\lambda t}) + y_0 e^{\lambda t}, \quad \lambda = -\frac{b}{m}, \quad \tilde{y} = \frac{p}{b} \tilde{u}$$

Examples

$$y(t) = \tilde{y} (1 - e^{\lambda t}) + y_0 e^{\lambda t}, \quad \tilde{y} = 1$$



Time-constant

Definition

$$\tau = \{t : y(t) = (1 - e^{-1}) \tilde{y} \approx 0.63 \tilde{y}\}$$

First-order systems

$$\tau = -\frac{1}{\lambda} \quad y(\tau) \approx 0.63 \tilde{y}, \quad y(3\tau) \approx 0.95 \tilde{y}$$

Rise-time

Definition

$$t_r = t_2 - t_1, \quad t_1 = \{t : y(t) = 0.1 \tilde{y}\}, \quad t_2 = \{t : y(t) = 0.9 \tilde{y}\}$$

First-order systems

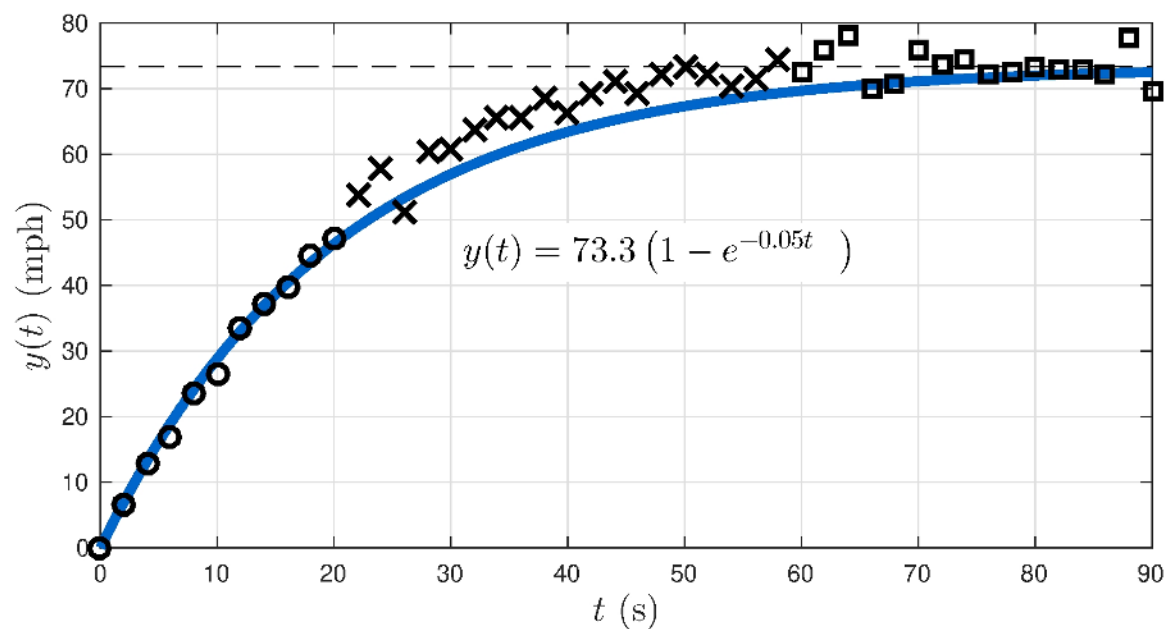
$$t_r = \ln(1/9) \lambda^{-1} = \ln(9) \tau \approx 2.2 \tau$$

2.4 Experimental Dynamic Response

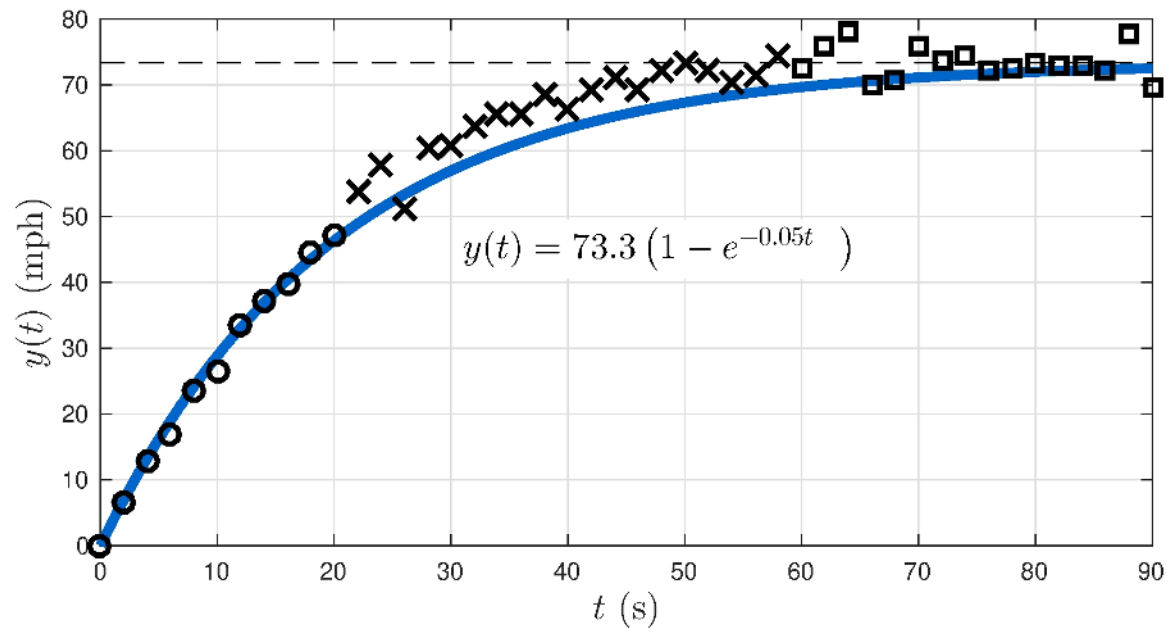
Experiment

Zero initial velocity

Constant pedal excursion: $\tilde{u} = 1$ in



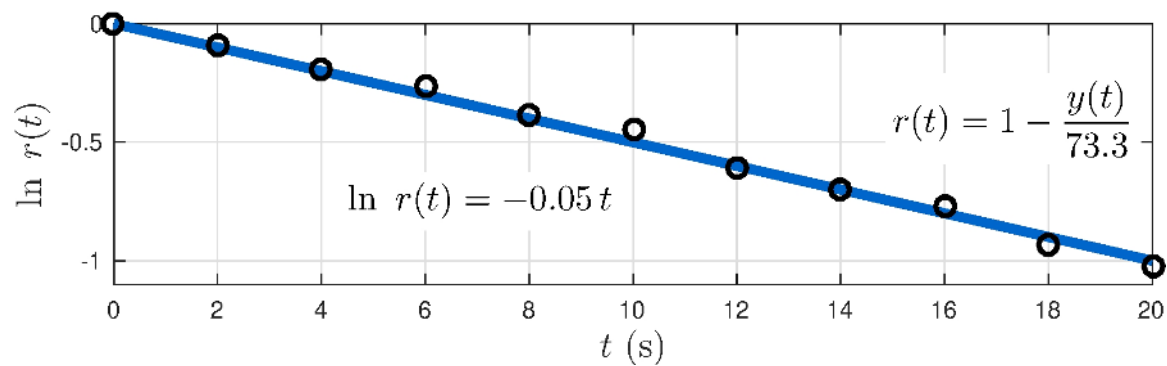
Model fitting



Squares: steady-state solution $\implies \tilde{y} \approx 73.3$ mph

Model fitting

$$r(t) = 1 - y(t)/\tilde{y} = e^{\lambda t} \quad \implies \quad \ln r(t) = \lambda t$$



Circles: $\lambda \approx -0.05 \text{ s}^{-1}$

Model fitting

$$\frac{b}{m} = -\lambda \approx 0.05 \text{ s}^{-1},$$

$$\frac{p}{b} = \frac{\tilde{y}}{\tilde{u}} \approx 73.3 \text{ mph/in},$$

$$\frac{p}{m} = \frac{b}{m} \times \frac{p}{b} = -\lambda \times \frac{\tilde{y}}{\tilde{u}} = 3.7 \text{ mph/(in s)}$$

Experimental model

$$\dot{y}(t) + 0.05 y(t) = 3.7 u(t)$$

Be careful

Experiment was conducted for $u(t) = \tilde{u}$ constant but we *wish* model to hold for general $u(t)$!

2.5 Dynamic Feedback Control

Car model

Dynamic model

$$\dot{y}(t) + \frac{b}{m} y(t) = \frac{p}{m} u(t)$$

Feedback controller

$$u(t) = K e(t), \quad e(t) = \bar{y} - y(t)$$

Closed-loop system

$$\dot{y}(t) + \left(\frac{b}{m} + \frac{p}{m} K \right) y(t) = \frac{p}{m} K \bar{y}$$

Closed-loop response

Dynamic model

$$\dot{y}(t) + \left(\frac{b}{m} + \frac{p}{m}K \right) y(t) = \frac{p}{m}K\bar{y}, \quad y(0) = y_0, \quad \lambda = -\frac{b}{m} - \frac{p}{m}K$$

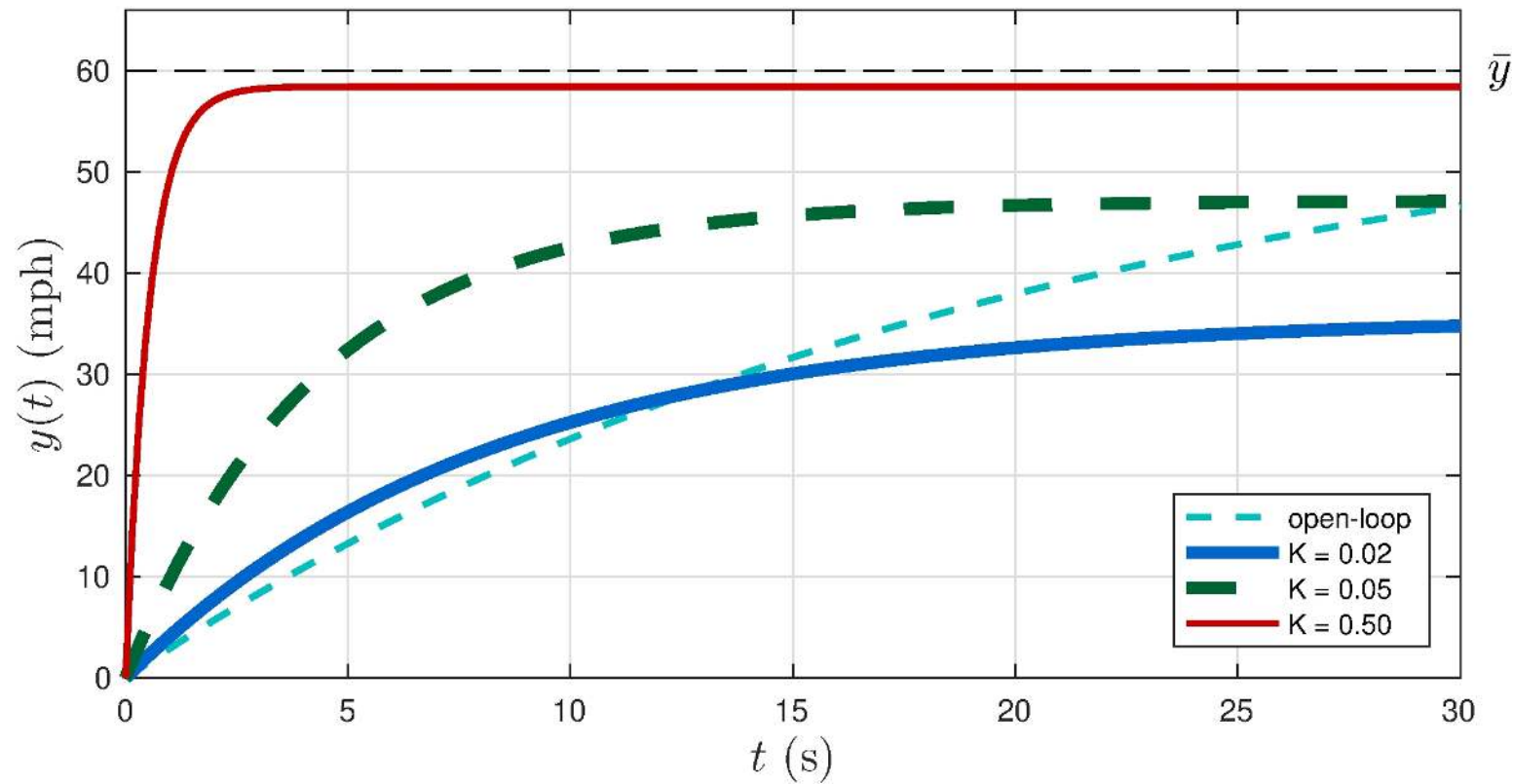
Output

$$y(t) = \tilde{y} (1 - e^{\lambda t}) + y_0 e^{\lambda t}, \quad \tilde{y} = \frac{\frac{p}{m}K}{\frac{b}{m} + \frac{p}{m}K} \bar{y} = \frac{\frac{p}{b}K}{1 + \frac{p}{b}K} \bar{y}$$

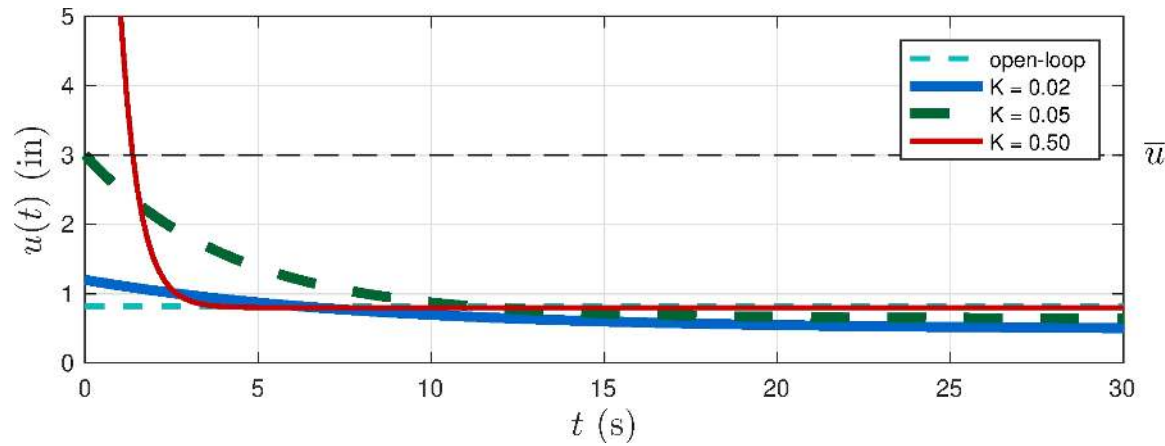
Time-constant

$$\lambda = -\frac{b}{m} - \frac{p}{m}K, \quad \tau = -\lambda^{-1} = \frac{m}{b + pK}$$

Output



Control



Largest control at $t = 0$

$$u(0) = Ke(0) = K(\bar{y} - y(0)) = K \bar{y}$$

Saturation

$$\bar{y} = 60 \text{ mph} \implies u(t) > \bar{u} = 3 \text{ in for } K > 0.05!$$

Steady-state

$$\lim_{t \rightarrow \infty} y(t) = \tilde{y} = H(0) \bar{y}, \quad H(0) = \frac{\frac{p}{b} K}{1 + \frac{p}{b} K}$$

$$\lim_{t \rightarrow \infty} e(t) = \bar{y} - \tilde{y} = S(0) \bar{y}, \quad S(0) = \frac{1}{1 + \frac{p}{b} K}$$

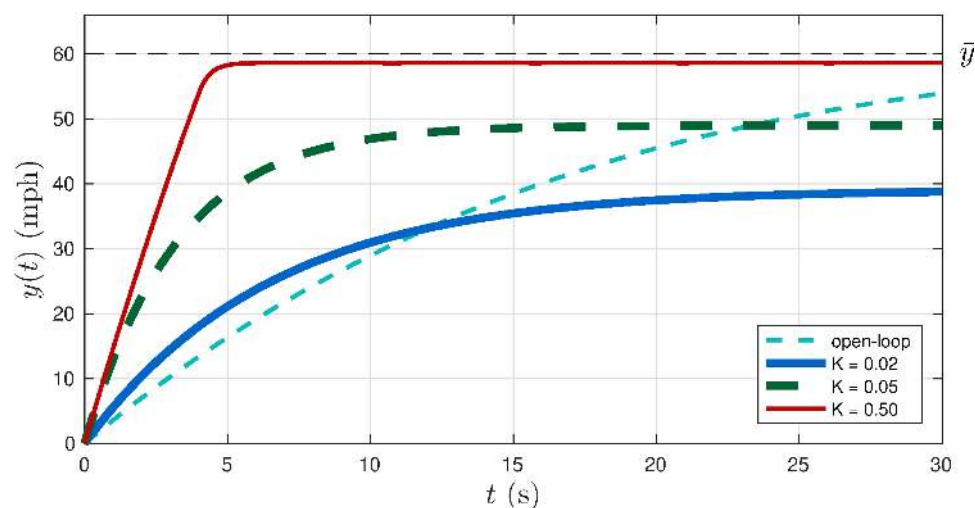
Effect of the gain

K	$H(0)$	$S(0)$	$\tilde{y}$ (mph)	τ (s)	t_r (s)
open-loop	1.00	0.00	60	20.0	43.9
0.02	0.60	0.40	36	8.1	17.8
0.05	0.79	0.21	47	4.3	9.4
0.50	0.97	0.03	58	0.5	1.2

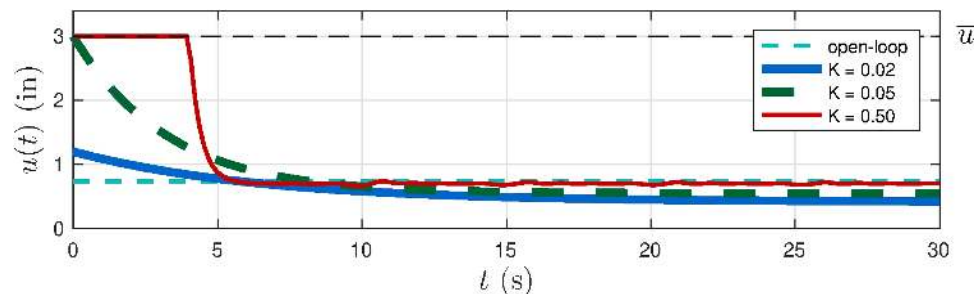
$$\bar{y} = 60 \text{ mph}$$

2.6 Nonlinear Models

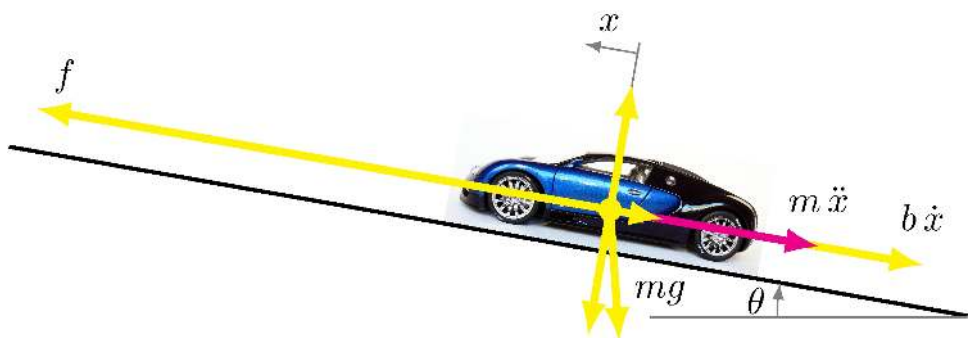
Output is rate-limited



Control saturates



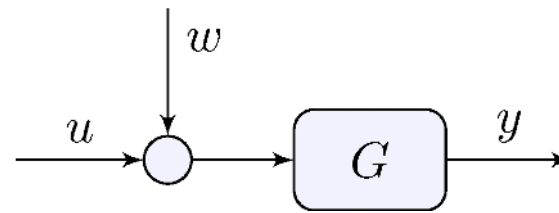
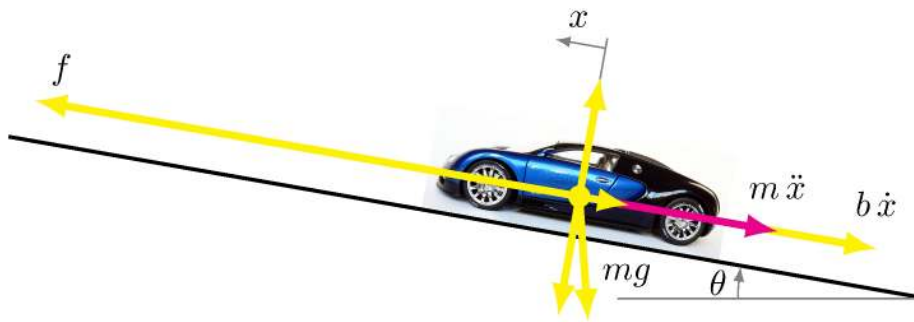
2.7 Disturbance Rejection



Car on inclined plane model

$$\dot{y}(t) + \frac{b}{m} y(t) = \frac{p}{m} u(t) - g \sin(\theta(t))$$

Disturbance Model



Car on inclined plane with disturbance

$$\dot{y}(t) + \frac{b}{m} y(t) = \frac{p}{m} (u(t) + w(t)), \quad w(t) = -\frac{mg}{p} \sin(\theta(t))$$

Car with disturbance

$$\dot{y}(t) + \frac{b}{m} y(t) = \frac{p}{m} (u(t) + w(t))$$

Constant disturbance

$$w(t) = \bar{w}, \quad t \geq 0$$

Feedback controller

$$u(t) = K e(t), \quad e(t) = \bar{y} - y(t)$$

Impact of disturbance on steady-state

Model

$$\dot{y}(t) + \left(\frac{b}{m} + \frac{p}{m} K \right) y(t) = \frac{p}{m} (K \bar{y} + \bar{w}), \quad y_0 = H(0) \bar{y}$$

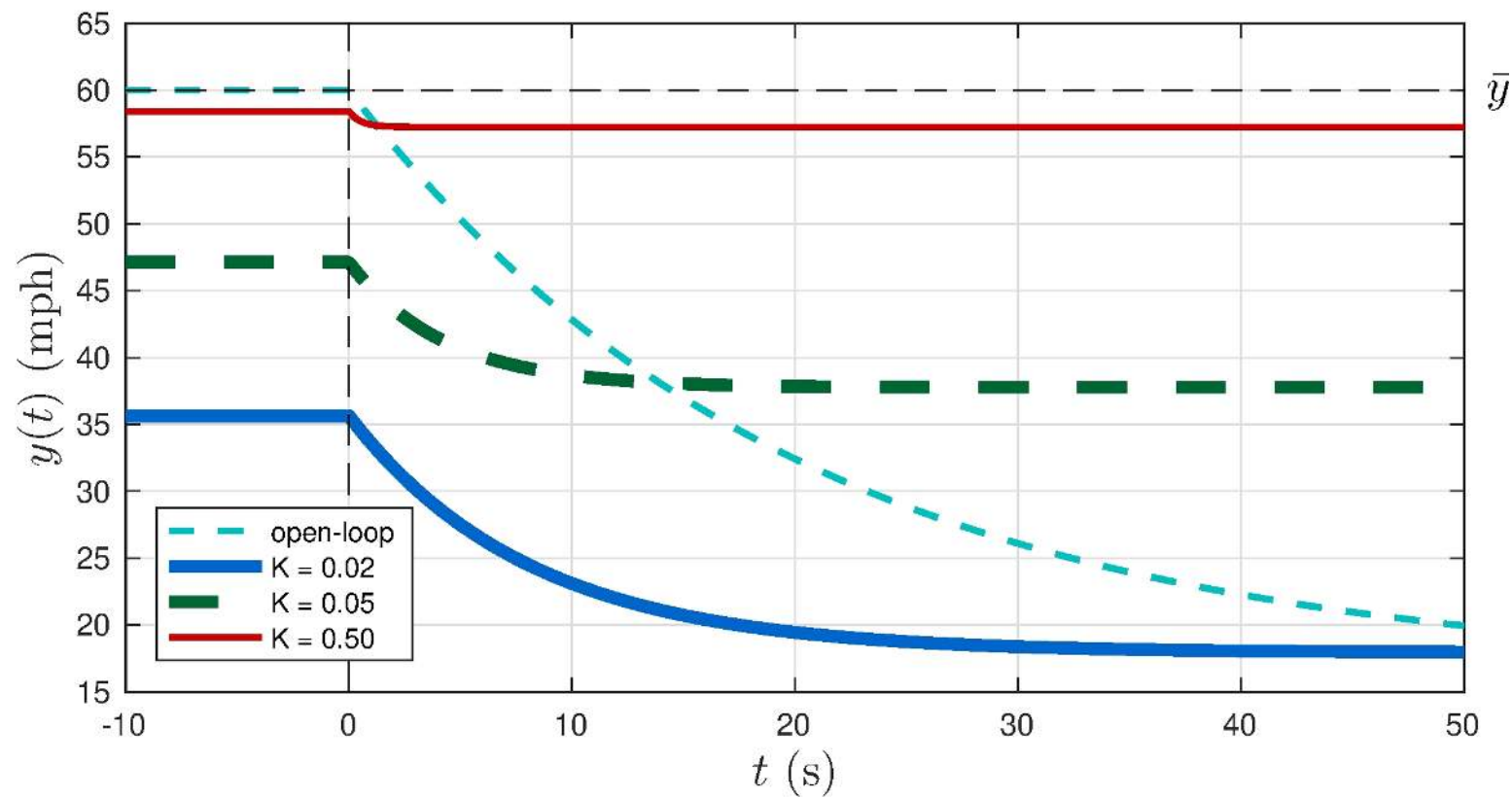
Output

$$y(t) = H(0) \bar{y} + (1 - e^{\lambda t}) D(0) \bar{w}, \quad H(0) = \frac{\frac{p}{b} K}{1 + \frac{p}{b} K}, \quad D(0) = \frac{p}{b} \frac{1}{1 + \frac{p}{b} K}$$

Deviation from steady-state

$$\Delta y(t) = y(t) - H(0) \bar{y} = (1 - e^{\lambda t}) D(0) \bar{w}$$

Output



2.8 Integral Action

Toilet tank model

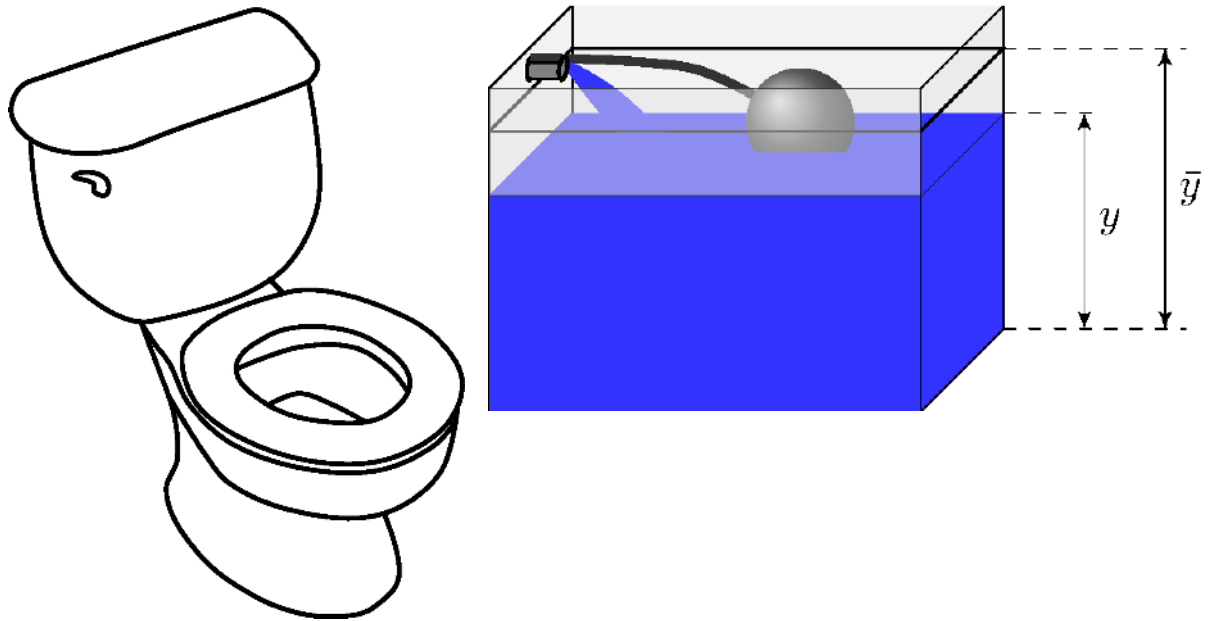
$$\dot{y}(t) = \frac{1}{A}u(t)$$

Ballcock valve

$$u(t) = K(\bar{y} - y)$$

Closed-loop model

$$\dot{y}(t) + \frac{K}{A}y(t) = \frac{K}{A}\bar{y}(t)$$



Closed-loop response

Model

$$\dot{y}(t) + \frac{K}{A}y(t) = \frac{K}{A}\bar{y}(t), \quad y(0) = y_0, \quad \lambda = -\frac{K}{A}$$

Output

$$y(t) = \tilde{y} (1 - e^{\lambda t}) + y_0 e^{\lambda t}, \quad \tilde{y} = \bar{y}$$

Steady-state

$$\begin{aligned} \lim_{t \rightarrow \infty} y(t) &= \tilde{y} = H(0) \bar{y}, & H(0) &= 1 \\ \lim_{t \rightarrow \infty} e(t) &= \bar{y} - \tilde{y} = S(0) \bar{y}, & S(0) &= 0 \end{aligned}$$

Integral action $\implies$ zero tracking error!

Can we replicate that result in a more general setting? Answer is in Chapter 4

Learn more

- [Book website](#)